Binomial Dist.

10/14

Counting the number of green lights I hit on my way to work out of 5.

Assumptions: 1) First light 50/50.

2) $G_2 \rightarrow G$ is 90%.

3) $R \rightarrow$ 50/50. R/G.

$P(GGGGR) = 0.5 \times 0.9 \times 0.9 \times 0.9 \times 0.1$

Real Assump: 1) each traffic light is independent.

2) all lights have the same prob of success. (getting a green light) is p.. $P(G) = p$.

$$P(GGGGR) = \underbrace{p \times p \times p \times p \times (1-p)}$$

multiply because ind events.

$$= p^4(1-p)$$

GRGGG, RGGGG, GGRGG, GGGRG.

$$P(\overbrace{4}^{\text{no. of success}} \text{ green lights out of } 5)$$

$$= 5 \times p^4(1-p).$$

5 ↑ combination term.

X = no. of green lights out 5.

$X \sim \text{Binom}(n, p)$ ← parameters

X follows

where, n = no. of trials.

p = Prob of success.

Binomial dist satisfies 4 main assumptions:

B : Binary outcomes in each trial.

I : Independent trials.

N : No of trials is fixed.

S : Same probability of success in each trial.

$$P(X=k) = {}^{n}C_{k} \times p^{k} \times (1-p)^{n-k}$$

4 success, 1 fail

$$P(X=4) = P\begin{pmatrix} SSSSF \\ \vdots \\ FSSSS \end{pmatrix} = {}^{5}C_{4} \times p \times p \times p \times p \times (1-p)$$

$k=4$, 5C4

↳ combination term.

$n C_k$ = n choose k (R func: choose(n,k))

$n! = n(n-1)(n-2)\cdots 1$

$$n C_k = \frac{n!}{(n-k)!\,k!}$$

$$5C_4 = \frac{5!}{(5-4)!\,4!} = \frac{5!}{1!\,4!}$$

$$= \frac{5 \times 4 \times 3 \times 2 \times 1}{1 \times 4 \times 3 \times 2 \times 1}$$

$$= 5$$

Binomial Dist:

No. of G I hit on my way to work out of 5.

$P(GGGGR)$

Realistic Assumptions: 1) R/G → 50/50 once I start from home.

2) G → G is 70%.

3) R → G is 50%.

$P(A \cap B) = P(A)P(B)$ ⇒ A & B is ind.

$\rightarrow P(GGGGR) = 0.3 \times 0.7 \times 0.7 \times 0.7 \times 0.3$

Ideal Assumption : 1) all traffic lights are independent.

2) Prob of success (getting a G) is constant for all traffic signals & is p. $(0 \le p \le 1)$

$P(GGGGR) = p \times p \times p \times p \times (1-p)$

$P(X = 4) = P(GGGGR) +$
$P(GRGGG) +$
$P(GGRGG) +$
$P(RGGGG) + P(GGGRG)$

$= 5p^4(1-p)$

combination term.

$P(X=k) = {}^5C_k\, p^k (1-p)^{n-k}$.

$P(RRRRR) = P(X=0) = {}^5C_0\, p^0 (1-p)^5$

$P(\overset{X=4}{\text{GGGGR}}) = 5p^4(1-p)$

$P(X=4, n=7$

GGGGRRR$)$

X = no. of success.

$X \sim$ Binomial dist.

$\Downarrow$

follows.

parameters of the binomial dist.

$X \sim \text{Binom}(n, p)$

where, n = num. of trials.

p = prob of success.

Assumptions:

B : Binary outcomes for each trial

I : Independent trials.

N : Number of trials is fixed

S : Same prob of success in each trial (denoted as p).

(Prob of failure denoted as q).

3 coin flips.

actual set of outcome $\{$ HHH, HHT, HTH, THH, TTH, THT, HTT, TTT $\}$.

X = no. of heads I get in 3 coin flips.

sample space for r.v. $\rightarrow \Omega_X = \{0, 1, 2, 3\}$

$X \sim \text{binom}(3, 0.5)$.

$$P(X=2) = {}_3C_2 (0.5)^2 (0.5)^1$$